

1. Strongly Connected Components

(a)

C_n : $\forall u, v \in V$: following the cycle gives a directed $u \rightarrow v$ path and a directed $v \rightarrow u$ path, so $u \sim v$. $1 \text{ SCC} = V$

P_n : Edges $1 \rightarrow 2 \rightarrow \dots \rightarrow n$. $\forall i < j$: directed path $i \rightarrow j$ exists, but no directed path $j \rightarrow i$ (all edges increase index), so $i \not\sim j$. $n \text{ SCCs: } \{1\}, \{2\}, \dots, \{n\}$

(b)

$V = \{1, \dots, n\}$. Place a directed cycle on $\{1, \dots, n - k + 1\}$:

$$E = \{i \rightarrow i+1 : 1 \leq i \leq n-k\} \cup \{(n-k+1) \rightarrow 1\}$$

Vertices $\{n-k+2, \dots, n\}$ are isolated.

The cycle forms 1 SCC. Each isolated vertex is a singleton SCC. $1 + (k - 1) = k \text{ SCCs}$

2. Graph Coloring

(a)

($\Rightarrow$) Let $c : V \rightarrow \{1, 2\}$ be a proper 2-coloring. Set $A = c^{-1}(1)$, $B = c^{-1}(2)$. $\forall uv \in E$: $c(u) \neq c(v)$, so one endpoint is in A and the other in $B \Rightarrow (A, B)$ is a bipartition.

($\Leftarrow$) Given bipartition (A, B) , define $c(v) = 1$ if $v \in A$, $c(v) = 2$ if $v \in B$. $\forall uv \in E$: endpoints lie in different parts $\Rightarrow c(u) \neq c(v)$. $\square$

(b)

Process vertices in any order $v_1, \dots, v_n$. Assign $c(v_i) = \min\{j \in \{1, \dots, \Delta+1\} : j \neq c(v_\ell) \forall v_\ell \in N(v_i), \ell < i\}$.

When coloring v_i : at most $\deg(v_i) \leq \Delta$ neighbors already colored, blocking $\leq \Delta$ colors from $\{1, \dots, \Delta+1\}$. Since $\Delta + 1 > \Delta$, at least one color is available. $\square$

(c)

$G_\Delta = K_{\Delta+1}$. Every vertex has degree Δ . All $\Delta + 1$ vertices are pairwise adjacent $\Rightarrow$ any proper coloring assigns distinct colors to all vertices $\Rightarrow$ needs $\Delta + 1$ colors. $\text{Not } \Delta\text{-colorable}$

3. Shortest Path Trees

(a)

Yes. Triangle: $w(u, w) = 1$, $w(w, v) = 1$, $w(u, v) = 2$.

Path $u-v$ (direct): weight 2. Path $u-w-v$: weight $1 + 1 = 2$. Two distinct shortest $u-v$ paths

(b)

$V = \{u, a, b, c\}$. Edges: $(u, a) = 3$, $(u, b) = 1$, $(b, a) = 2$, $(a, c) = 1$.

v	u	b	a	c
$d(u, v)$	0	1	3	4

Two shortest paths to a : $u-a$ (weight 3) and $u-b-a$ (weight $1 + 2 = 3$).

Choose $P(u, b) = u-b$, $P(u, a) = u-a$, $P(u, c) = u-b-a-c$.

$$T(u) = \{(u, a), (u, b), (b, a), (a, c)\} \quad (4 \text{ edges, } 4 \text{ vertices}) \implies \boxed{\text{cycle } u-a-b-u}$$

(c)

For each $v \neq u$, choose predecessor $\pi(v)$ satisfying $d(u, v) = d(u, \pi(v)) + w(\pi(v), v)$. Define $P(u, v) = P(u, \pi(v)) \cdot (\pi(v), v)$.

$T(u)$ is a tree:

1. $n - 1$ **edges**: Each $v \neq u$ contributes exactly one edge $(\pi(v), v)$. Distinct v yield distinct edges.
2. **Connected**: The chain $v, \pi(v), \pi(\pi(v)), \dots$ terminates at u , since $w(e) > 0 \implies d(u, \pi(v)) < d(u, v)$ (strictly decreasing). Every edge in this chain belongs to $T(u)$, so u reaches every v .
3. Connected + n vertices + $n - 1$ edges $\implies$ tree. $\square$

4. Shortest Shortest-Path Tree

Counterexample. $V = \{a, b, c, d\}$.

edge	(a, b)	(b, c)	(c, d)	(a, c)	(b, d)
w	2	2	2	3	3

MST (Kruskal): $\{(a, b), (b, c), (c, d)\}$, weight = $2 + 2 + 2 = 6$.

Best SPT from each root:

u	$T(u)$	$W(u)$
a	$(a, b) = 2, (a, c) = 3, (c, d) = 2$	7
b	$(b, a) = 2, (b, c) = 2, (b, d) = 3$	7
c	$(c, a) = 3, (c, b) = 2, (c, d) = 2$	7
d	$(d, c) = 2, (d, b) = 3, (b, a) = 2$	7

$d(a, d) = 5 < 6 = w(a-b-c-d)$, so every SPT from a or d must use a weight-3 shortcut. For roots b and c , the shortest path to the far vertex also requires a weight-3 edge.

$W(u^*) = 7 > 6 = W(\text{MST}) \implies$ Dylan's algorithm does not find an MST